Year 7

REVISION PACK

Unit 4

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 4 · Fractions of amounts

Q	Working	Answer
simplifying and fractions of amounts		
1.1	**(a)** $\div 5: \ \frac{15}{25} = \frac{3}{5}$ **(b)** $145 \div 5 = 29$ $29 \times 3 = 87$ **(c)** $48 \div 4 = 12$ $12 \times 3 = 36$ **(d)** $21 \div 7 = 3$ $3 \times 5 = 15$	**(a)** $\frac{3}{5}$ **(b) \$87** **(c) 36** **(d) 15**
1.2	**(a)** $\div 8: \ \frac{16}{24} = \frac{2}{3}$ **(b)** $117 \div 9 = 13$ $13 \times 4 = 52$ **(c)** $35 \div 5 = 7$ $7 \times 2 = 14$ **(d)** $24 \div 8 = 3$ $3 \times 3 = 9$	**(a)** $\frac{2}{3}$ **(b) \$52** **(c) 14** **(d) 9**
1.3	**(a)** $\div 6: \ \frac{12}{42} = \frac{2}{7}$ **(b)** $162 \div 6 = 27$ $27 \times 5 = 135$ **(c)** $63 \div 7 = 9$ $9 \times 4 = 36$ **(d)** $30 \div 5 = 6$ $6 \times 2 = 12$	**(a)** $\frac{2}{7}$ **(b) \$135** **(c) 36** **(d) 12**

1.4	**(a)** $\div 9: \ \frac{27}{36} = \frac{3}{4}$ **(b)** $216 \div 8 = 27$ $27 \times 3 = 81$ **(c)** $72 \div 9 = 8$ $8 \times 5 = 40$ **(d)** $36 \div 12 = 3$ $3 \times 7 = 21$	**(a)** $\frac{3}{4}$ **(b) \$81** **(c) 40** **(d) 21**
1.5	**(a)** $\div 7: \ \frac{14}{49} = \frac{2}{7}$ **(b)** $230 \div 10 = 23$ $23 \times 7 = 161$ **(c)** $64 \div 8 = 8$ $8 \times 3 = 24$ **(d)** $45 \div 9 = 5$ $5 \times 4 = 20$	**(a)** $\frac{2}{7}$ **(b) \$161** **(c) 24** **(d) 20**
1.6	**(a)** $\div 15: \ \frac{30}{45} = \frac{2}{3}$ **(b)** $138 \div 3 = 46$ $46 \times 2 = 92$ **(c)** $54 \div 6 = 9$ $9 \times 5 = 45$ **(d)** $40 \div 10 = 4$ $4 \times 3 = 12$	**(a)** $\frac{2}{3}$ **(b) \$92** **(c) 45** **(d) 12**

Unit 4 · Fractions and decimals

Q	Working	Answer
fractions as decimals and mixed numbers		
2.1	**(a)** $3 \times 0.2 = 0.6$ **(b)** $0.2 \div 2 = 0.1$ **(c)** $11 \div 4 = 2 \text{ r } 3$ **(d)** $19 \div 6 = 3 \text{ r } 1$	**(a) 0.6** **(b) 0.1** **(c)** $2\frac{3}{4}$ **(d)** $3\frac{1}{6}$

2.2	**(a)** $5 \times 0.125 = 0.625$ **(b)** $0.125 \div 2 = 0.0625$ **(c)** $13 \div 3 = 4 \text{ r } 1$ **(d)** $23 \div 5 = 4 \text{ r } 3$	**(a) 0.625** **(b) 0.0625** **(c)** $4\frac{1}{3}$ **(d)** $4\frac{3}{5}$
2.3	**(a)** $7 \times 0.05 = 0.35$ **(b)** $0.05 \div 2 = 0.025$ **(c)** $15 \div 2 = 7 \text{ r } 1$ **(d)** $29 \div 8 = 3 \text{ r } 5$	**(a) 0.35** **(b) 0.025** **(c)** $7\frac{1}{2}$ **(d)** $3\frac{5}{8}$
2.4	**(a)** $9 \times 0.1 = 0.9$ **(b)** $0.1 \div 2 = 0.05$ **(c)** $17 \div 6 = 2 \text{ r } 5$ **(d)** $22 \div 7 = 3 \text{ r } 1$	**(a) 0.9** **(b) 0.05** **(c)** $2\frac{5}{6}$ **(d)** $3\frac{1}{7}$
2.5	**(a)** $3 \times 0.04 = 0.12$ **(b)** $0.04 \div 2 = 0.02$ **(c)** $31 \div 9 = 3 \text{ r } 4$ **(d)** $27 \div 4 = 6 \text{ r } 3$	**(a) 0.12** **(b) 0.02** **(c)** $3\frac{4}{9}$ **(d)** $6\frac{3}{4}$
2.6	**(a)** $3 \times 0.5 = 1.5$ **(b)** $0.5 \div 2 = 0.25$ **(c)** $25 \div 3 = 8 \text{ r } 1$ **(d)** $37 \div 10 = 3 \text{ r } 7$	**(a) 1.5** **(b) 0.25** **(c)** $8\frac{1}{3}$ **(d)** $3\frac{7}{10}$

Unit 4 · Fractions, decimals and percentages

Q	Working	Answer
completing a fraction, decimal and percentage table		
3.1	$75\% = \frac{75}{100} = 0.75$ $160\% = \frac{160}{100} = 1.6$ $\frac{2}{5} = \frac{40}{100} = 0.4$	**75% = 0.75 =** $\frac{3}{4}$ **160% = 1.6 =** $1\frac{3}{5}$ $\frac{2}{5}$ **= 0.4 = 40%**

3.2	$45\% = \frac{45}{100} = 0.45$ $120\% = \frac{120}{100} = 1.2$ $\frac{3}{4} = \frac{75}{100} = 0.75$	**45% = 0.45 =** $\frac{9}{20}$ **120% = 1.2 =** $1\frac{1}{5}$ $\frac{3}{4}$ **= 0.75 = 75%**
3.3	$80\% = \frac{80}{100} = 0.8$ $250\% = \frac{250}{100} = 2.5$ $\frac{7}{20} = \frac{35}{100} = 0.35$	**80% = 0.8 =** $\frac{4}{5}$ **250% = 2.5 =** $2\frac{1}{2}$ $\frac{7}{20}$ **= 0.35 = 35%**
3.4	$15\% = \frac{15}{100} = 0.15$ $130\% = \frac{130}{100} = 1.3$ $\frac{9}{10} = \frac{90}{100} = 0.9$	**15% = 0.15 =** $\frac{3}{20}$ **130% = 1.3 =** $1\frac{3}{10}$ $\frac{9}{10}$ **= 0.9 = 90%**
3.5	$65\% = \frac{65}{100} = 0.65$ $175\% = \frac{175}{100} = 1.75$ $\frac{1}{4} = \frac{25}{100} = 0.25$	**65% = 0.65 =** $\frac{13}{20}$ **175% = 1.75 =** $1\frac{3}{4}$ $\frac{1}{4}$ **= 0.25 = 25%**
3.6	$5\% = \frac{5}{100} = 0.05$ $110\% = \frac{110}{100} = 1.1$ $\frac{3}{25} = \frac{12}{100} = 0.12$	**5% = 0.05 =** $\frac{1}{20}$ **110% = 1.1 =** $1\frac{1}{10}$ $\frac{3}{25}$ **= 0.12 = 12%**

Unit 4 · Multiplying and comparing fractions

Q	Working	Answer
multiplying and comparing fractions		
4.1	**(a)** $\frac{2}{3} \times 7 = \frac{14}{3} = 4\frac{2}{3}$ **(b)** $\frac{3}{5} = \frac{9}{15}$, $\frac{2}{3} = \frac{10}{15}$ **(c)** $\frac{12}{36} = \frac{1}{3}$	**(a)** $4\frac{2}{3}$ **(b)** $\frac{2}{3}$ **(c) shown**

4.2	**(a)** $\frac{3}{5} \times 4 = \frac{12}{5} = 2\frac{2}{5}$ **(b)** $\frac{4}{5} = \frac{8}{10}$, $\frac{7}{10} = \frac{7}{10}$ **(c)** $\frac{6}{24} = \frac{1}{4}$	**(a)** $2\frac{2}{5}$ **(b)** $\frac{4}{5}$ **(c) shown**
4.3	**(a)** $\frac{1}{6} \times 9 = \frac{9}{6} = 1\frac{1}{2}$ **(b)** $\frac{5}{8} = \frac{15}{24}$, $\frac{2}{3} = \frac{16}{24}$ **(c)** $\frac{15}{60} = \frac{1}{4}$	**(a)** $1\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c) shown**
4.4	**(a)** $\frac{4}{7} \times 5 = \frac{20}{7} = 2\frac{6}{7}$ **(b)** $\frac{3}{4} = \frac{27}{36}$, $\frac{7}{9} = \frac{28}{36}$ **(c)** $\frac{20}{60} = \frac{1}{3}$	**(a)** $2\frac{6}{7}$ **(b)** $\frac{7}{9}$ **(c) shown**
4.5	**(a)** $\frac{2}{9} \times 6 = \frac{12}{9} = 1\frac{1}{3}$ **(b)** $\frac{5}{7} = \frac{15}{21}$, $\frac{2}{3} = \frac{14}{21}$ **(c)** $\frac{21}{63} = \frac{1}{3}$	**(a)** $1\frac{1}{3}$ **(b)** $\frac{5}{7}$ **(c) shown**
4.6	**(a)** $\frac{3}{8} \times 10 = \frac{30}{8} = 3\frac{3}{4}$ **(b)** $\frac{7}{12} = \frac{35}{60}$, $\frac{3}{5} = \frac{36}{60}$ **(c)** $\frac{42}{84} = \frac{1}{2}$	**(a)** $3\frac{3}{4}$ **(b)** $\frac{3}{5}$ **(c) shown**

Unit 4 · Adding and subtracting fractions

Q	Working	Answer
adding and subtracting fractions and mixed numbers		

5.1	**(a)** $\frac{3}{8}+\frac{2}{8}=\frac{5}{8}$ **(b)** $\frac{11}{12}-\frac{9}{12}=\frac{2}{12}=\frac{1}{6}$ **(c)** $\frac{21}{5}+\frac{27}{10}=\frac{42}{10}+\frac{27}{10}$ $=\frac{69}{10}=6\frac{9}{10}$ **(d)** $\frac{25}{4}-\frac{31}{12}=\frac{75}{12}-\frac{31}{12}$ $=\frac{44}{12}=3\frac{2}{3}$	**(a)** $\frac{5}{8}$ **(b)** $\frac{1}{6}$ **(c)** $6\frac{9}{10}$ **(d)** $3\frac{2}{3}$
5.2	**(a)** $\frac{1}{6}+\frac{4}{6}=\frac{5}{6}$ **(b)** $\frac{7}{10}-\frac{2}{10}=\frac{5}{10}=\frac{1}{2}$ **(c)** $\frac{11}{4}+\frac{3}{2}=\frac{11}{4}+\frac{6}{4}$ $=\frac{17}{4}=4\frac{1}{4}$ **(d)** $\frac{16}{3}-\frac{23}{6}=\frac{32}{6}-\frac{23}{6}$ $=\frac{9}{6}=1\frac{1}{2}$	**(a)** $\frac{5}{6}$ **(b)** $\frac{1}{2}$ **(c)** $4\frac{1}{4}$ **(d)** $1\frac{1}{2}$
5.3	**(a)** $\frac{2}{9}+\frac{3}{9}=\frac{5}{9}$ **(b)** $\frac{5}{8}-\frac{2}{8}=\frac{3}{8}$ **(c)** $\frac{17}{5}+\frac{13}{10}=\frac{34}{10}+\frac{13}{10}$ $=\frac{47}{10}=4\frac{7}{10}$ **(d)** $\frac{25}{6}-\frac{5}{3}=\frac{25}{6}-\frac{10}{6}$ $=\frac{15}{6}=2\frac{1}{2}$	**(a)** $\frac{5}{9}$ **(b)** $\frac{3}{8}$ **(c)** $4\frac{7}{10}$ **(d)** $2\frac{1}{2}$

5.4	**(a)** $\frac{2}{8}+\frac{3}{8}=\frac{5}{8}$ **(b)** $\frac{9}{10}-\frac{4}{10}=\frac{5}{10}=\frac{1}{2}$ **(c)** $\frac{11}{6}+\frac{10}{3}=\frac{11}{6}+\frac{20}{6}$ $=\frac{31}{6}=5\frac{1}{6}$ **(d)** $\frac{15}{2}-\frac{24}{5}=\frac{75}{10}-\frac{48}{10}$ $=\frac{27}{10}=2\frac{7}{10}$	**(a)** $\frac{5}{8}$ **(b)** $\frac{1}{2}$ **(c)** $5\frac{1}{6}$ **(d)** $2\frac{7}{10}$
5.5	**(a)** $\frac{3}{10}+\frac{5}{10}=\frac{8}{10}=\frac{4}{5}$ **(b)** $\frac{10}{12}-\frac{3}{12}=\frac{7}{12}$ **(c)** $\frac{9}{4}+\frac{29}{8}=\frac{18}{8}+\frac{29}{8}$ $=\frac{47}{8}=5\frac{7}{8}$ **(d)** $\frac{32}{5}-\frac{11}{4}=\frac{128}{20}-\frac{55}{20}$ $=\frac{73}{20}=3\frac{13}{20}$	**(a)** $\frac{4}{5}$ **(b)** $\frac{7}{12}$ **(c)** $5\frac{7}{8}$ **(d)** $3\frac{13}{20}$
5.6	**(a)** $\frac{4}{12}+\frac{3}{12}=\frac{7}{12}$ **(b)** $\frac{12}{15}-\frac{5}{15}=\frac{7}{15}$ **(c)** $\frac{17}{3}+\frac{7}{6}=\frac{34}{6}+\frac{7}{6}$ $=\frac{41}{6}=6\frac{5}{6}$ **(d)** $\frac{25}{8}-\frac{3}{2}=\frac{25}{8}-\frac{12}{8}$ $=\frac{13}{8}=1\frac{5}{8}$	**(a)** $\frac{7}{12}$ **(b)** $\frac{7}{15}$ **(c)** $6\frac{5}{6}$ **(d)** $1\frac{5}{8}$

Unit 4 · Fraction problems

Q	Working	Answer
a fraction of a fraction		

6.1	**(a)** $\frac{1}{6} \times 768000 = 128000$ $\frac{3}{4} \times 128000 = 96000$ **(b)** $girls = 1 - \frac{3}{5} = \frac{2}{5}$ $\text{left-handed} = 1 - \frac{7}{8} = \frac{1}{8}$ $\frac{1}{8} \times \frac{2}{5} = \frac{1}{20}$	**(a) 96,000 km²** **(b)** $\frac{1}{20}$
6.2	**(a)** $\frac{1}{4} \times 4800 = 1200$ $\frac{2}{3} \times 1200 = 800$ **(b)** $boys = 1 - \frac{1}{2} = \frac{1}{2}$ $\text{left-handed} = 1 - \frac{4}{5} = \frac{1}{5}$ $\frac{1}{5} \times \frac{1}{2} = \frac{1}{10}$	**(a) 800 m²** **(b)** $\frac{1}{10}$
6.3	**(a)** $\frac{2}{3} \times 36000 = 24000$ $\frac{1}{4} \times 24000 = 6000$ **(b)** $girls = 1 - \frac{3}{7} = \frac{4}{7}$ $\text{left-handed} = 1 - \frac{5}{6} = \frac{1}{6}$ $\frac{1}{6} \times \frac{4}{7} = \frac{2}{21}$	**(a) 6,000 km²** **(b)** $\frac{2}{21}$
6.4	**(a)** $\frac{2}{5} \times 1500 = 600$ $\frac{5}{6} \times 600 = 500$ **(b)** $boys = 1 - \frac{5}{9} = \frac{4}{9}$ $\text{left-handed} = 1 - \frac{2}{3} = \frac{1}{3}$ $\frac{1}{3} \times \frac{4}{9} = \frac{4}{27}$	**(a) 500 m²** **(b)** $\frac{4}{27}$

6.5	**(a)** $\frac{3}{4} \times 90000 = 67500$ $\frac{2}{9} \times 67500 = 15000$ **(b)** $girls = 1 - \frac{1}{3} = \frac{2}{3}$ $\text{left-handed} = 1 - \frac{9}{10} = \frac{1}{10}$ $\frac{1}{10} \times \frac{2}{3} = \frac{1}{15}$	**(a) 15,000 km²** **(b)** $\frac{1}{15}$
6.6	**(a)** $\frac{5}{7} \times 6300 = 4500$ $\frac{3}{5} \times 4500 = 2700$ **(b)** $boys = 1 - \frac{4}{9} = \frac{5}{9}$ $\text{left-handed} = 1 - \frac{3}{4} = \frac{1}{4}$ $\frac{1}{4} \times \frac{5}{9} = \frac{5}{36}$	**(a) 2,700 m²** **(b)** $\frac{5}{36}$

Unit 4 · Fraction problems

Q	Working	Answer
problems with adding and multiplying fractions		
7.1	**(a)** $\frac{4}{5} + \frac{3}{20} = \frac{19}{20}$ $\frac{19}{20} - \frac{1}{2} = \frac{9}{20}$ **(b)** $\frac{13}{5} \times \frac{3}{4} = \frac{39}{20} = 1\frac{19}{20}$	**(a)** $\frac{9}{20}$ **of a bottle** **(b) No, the area is** $1\frac{19}{20}$ **cm²**
7.2	**(a)** $\frac{2}{3} + \frac{5}{6} = \frac{3}{2}$ $\frac{3}{2} - \frac{1}{3} = \frac{7}{6}$ **(b)** $\frac{5}{4} \times \frac{2}{5} = \frac{10}{20} = \frac{1}{2}$	**(a)** $1\frac{1}{6}$ **of a bottle** **(b) Yes, the area is** $\frac{1}{2}$ **cm²**
7.3	**(a)** $\frac{7}{8} + \frac{1}{4} = \frac{9}{8}$ $\frac{9}{8} - \frac{3}{8} = \frac{3}{4}$ **(b)** $\frac{10}{3} \times \frac{3}{5} = \frac{30}{15} = 2$	**(a)** $\frac{3}{4}$ **of a bottle** **(b) No, the area is** 2 **cm²**

7.4	**(a)** $\frac{3}{5}+\frac{7}{10}=\frac{13}{10}$ $\frac{13}{10}-\frac{1}{2}=\frac{4}{5}$ **(b)** $\frac{5}{2}\times\frac{4}{5}=\frac{20}{10}=2$	**(a)** $\frac{4}{5}$ **of a bottle** **(b) No, the area is** 2 **cm²**
7.5	**(a)** $\frac{5}{6}+\frac{1}{3}=\frac{7}{6}$ $\frac{7}{6}-\frac{2}{3}=\frac{1}{2}$ **(b)** $\frac{7}{4}\times\frac{2}{7}=\frac{14}{28}=\frac{1}{2}$	**(a)** $\frac{1}{2}$ **of a bottle** **(b) Yes, the area is** $\frac{1}{2}$ **cm²**
7.6	**(a)** $\frac{9}{10}+\frac{2}{5}=\frac{13}{10}$ $\frac{13}{10}-\frac{3}{5}=\frac{7}{10}$ **(b)** $\frac{9}{2}\times\frac{2}{3}=\frac{18}{6}=3$	**(a)** $\frac{7}{10}$ **of a bottle** **(b) No, the area is** 3 **cm²**